

Bài 1:

```
interface Mail {  
    void sendMail();  
    void log();  
}  
  
class MailImpl implements Mail {  
    @Override  
    void sendMail() {  
        System.out.println("Dang Dang gửi mail...");  
    }  
  
    @Override  
    void log() {  
        System.out.println("Dang ghi log... tang tang nhap");  
    }  
}  
  
class Main {  
    public static void main(String[] args) {  
        MailImpl e = new MailImpl();  
        e.sendMail();  
        e.log();  
    }  
}
```


Bài 2:

```
interface Shape {  
    double getArea();  
    double getPerimeter();  
    void draw();  
}
```

```
class Rectangle implements Shape {
```

```
    @Override
```

```
    double length;
```

```
    private double length;
```

```
    private double width;
```

```
    public Rectangle (double length, double width) {
```

```
        this.length = length;
```

```
        this.width = width;
```

```
    }
```

```
    @Override
```

```
    public double getArea() {
```

```
        return return area * rong length * width;
```

```
    }
```


@Override

```
public double getPerimeter() {  
    return (length + width) * 2;  
}
```

@Override

```
public void draw() {  
    System.out.println("Ve HCN");  
}  
}
```

```
class Circle implements Shape {  
    private double radius;  
    public Circle(double radius) {  
        this.radius = radius;  
    }  
}
```

@Override

```
public double getArea() {  
    return Math.PI * radius * radius;  
}
```

@Override

trăng tình khôi đoạt khôi nguyên


```
public double getPerimeter () {  
    return 2 * Math.PI * radius;  
}
```

```
@Override
```

```
public void draw () {  
    System.out.println ("Vẽ hình tròn");  
}
```

```
}
```

```
class Main {
```

```
    public static void main (String [] args) {
```

```
        Shape h1 = new Rectangle (5, 3);
```

```
        Shape h2 = new Circle (2);
```

```
        h1.draw ();
```

```
        System.out.println ("Diện tích: " + h1.getArea());
```

```
        System.out.println ("Chu vi: " + h1.getPerimeter());
```

```
        h2.draw ();
```

```
        System.out.println ("Diện tích: " + h2.getArea());
```

```
        System.out.println ("Chu vi: " + h2.getPerimeter());
```

```
    }
```

```
}
```


Bài 3:

```
class HangSanXuat {  
    private String tenHangSanXuat;  
    private String tenQuocGia;  
    public HangSanXuat (String tenHangSanXuat, String tenQuocGia)  
    {  
        this.tenHangSanXuat = tenHangSanXuat;  
        this.tenQuocGia = tenQuocGia;  
    }  
    public String getTenHangSanXuat () {  
        return tenHangSanXuat;  
    }  
}  
  
abstract class PhuongTienDiChuyen {  
    protect String loaiPhuongTien;  
    protect HangSanXuat hangSanXuat;  
    public PhuongTienDiChuyen (String loaiPhuongTien,  
    HangSanXuat hangSanXuat) {  
        this.loaiPhuongTien = loaiPhuongTien;  
        this.hangSanXuat = hangSanXuat;  
    }  
  
    public String layTenHangSanXuat () {  
        return hangSanXuat.getTenHangSanXuat ();  
    }  
}
```




```

public void batDau () {
    System.out.println("Xe dang bat dau ...");
}

public void tangToc () {
    System.out.println("Xe tang toc dan ...");
}

public void dungLai () {
    System.out.println("Xe dung");
}

abstract double layVanToc();
}

```

```

class XeDap extends PhuongTienDiChuyen {
    public XeDap (HangSanXuat hangSanXuat) {
        super("Xe dap", hangSanXuat);
    }

    @Override
    public double layVanToc () {
        return 20;
    }
}

```




```

class XeOto extends PhuongTienDiChuyen {
    public XeOto {
        private String loaiNhiemLieu;
        public XeOto (String HangSanXuat HangSanXuat hangSanXuat, String loaiNhiemLieu) {
            super ("Xe Oto", hangSanXuat);
            this.loaiNhiemLieu = loaiNhiemLieu;
        }
        @Override
        public double layCanh () {
            return 80;
        }
    }
}

```

```

class MayBay extends PhuongTienDiChuyen {
    private String loaiNhiemLieu;
    public MayBay (String HangSanXuat HangSanXuat hangSanXuat, String loaiNhiemLieu) {
        super ("May Bay", hangSanXuat);
        this.loaiNhiemLieu = loaiNhiemLieu;
    }
    void catCanh () {
        System.out.println ("May bay cat Canh");
    }
    void NohaCanh () {
        System.out.println ("May bay ha canh");
    }
}

```




```
double layVanToc () {  
    return 900;  
}
```

```
public class main {  
    public static void main (String[] args) {  
        HangSanXuat h1 = new HangSanXuat ("Honda", "Nhật Bản");  
        HangSanXuat h2 = new HangSanXuat ("Toyota", "Nhật Bản");  
        HangSanXuat h3 = new HangSanXuat ("Boeing", "Mỹ");  
  
        XeDap xeDap = new XeDap (h1);  
        XeOto xeOto = new XeOto (h2, "Xăng");  
        mayBay  
        Xe mayBay = new MayBay (h3, "Nhiên liệu máy bay");  
  
        System.out.println ("Xe đạp: " + xeDap.layTenHangSanXuat ()  
+ " ", "Van toc la: " + xeDap.layVanToc ());  
        System.out.println ("Xe oto: " + xeOto.layTenHangSanXuat ()  
+ " ", "Van toc la: " + xeOto.layVanToc ());  
        System.out.println ("May bay: " + mayBay.layTenHangSanXuat ()  
+ " ", "Van toc la: " + mayBay.layVanToc ());  
    }  
}
```



Bài 4 Thứ ngày tháng năm

```
public abstract class Car {  
    protected String color;  
    protected double speed;  
  
    public Car (String color, double speed) {  
        this.color = color;  
        this.speed = speed;  
    }  
  
    public abstract void accelerate ();  
    public String getColor () {  
        return color;  
    }  
  
    public double getSpeed () {  
        return speed;  
    }  
}
```

```
public class Bmw extends Car {  
    public Bmw (String color, double speed) {  
        super (color, speed);  
    }  
}
```


@Override

public void accelerate () {

 System.out.println ("BMW đang tăng tốc");

}

@Override

public String toString () {

 return "BMW - Color: " + getColor () +
 ", Speed: " + getSpeed ();

}

}

class Civic extends Car {

 public Civic (String color, double speed) {

 super (color, speed);

}

@Override

public void accelerate () {

 System.out.println ("Civic đang tăng tốc");

}

@Override

public String toString () {

 return "Civic - Color: " + getColor () + ", Speed: "


```
+ getSpeed ();
```

```
}
```

```
}
```

```
public class Main {
```

```
    public static void main (String [] args) {
```

```
        Car c1c1 = new Bmw ("Black", 100);
```

```
        Car c2 = new Civic ("Red", 80);
```

```
        c1.accelerate();
```

```
        System.out.println (car 1);
```

```
        c2.accelerate();
```

```
        System.out.println (Car 2);
```

```
}
```

```
}
```

Bài 5:

```
Interface moveable {
```

```
    void moveUp ();
```

```
    void moveDown ();
```

```
    void moveLeft ();
```

```
    void moveRight ();
```

```
}
```


class MoveablePoint implements Moveable {

int x;

int y;

int xSpeed;

int ySpeed;

public MoveablePoint(int x, int y, int xSpeed,
int ySpeed) {

 this.x = x;

 this.y = y;

 this.xSpeed = xSpeed;

 this.ySpeed = ySpeed;

}

@Override

void moveUp() {

 y -= ySpeed;

}

@Override moveDown() {

 y += ySpeed;

}

@Override


```
public void moveLeft() {
```

```
    x -= xSpeed;
```

```
}
```

```
@Override
```

```
void moveRight() {
```

```
    x += xSpeed;
```

```
}
```

```
@Override
```

```
public String toString() {
```

```
    return "Point (" + x + ", " + y + ") speed = (" +  
    xSpeed + ", " + ySpeed + ")";
```

```
}
```

```
}
```

```
public class MoveableCircle implements Moveable {
```

```
    int radius;
```

```
    MoveablePoint center;
```

```
    public MoveableCircle (MoveablePoint center,  
        int radius) {
```

```
        this.center = center;
```

```
        this.radius = radius;
```

```
}
```


Thứ ngày tháng năm

@Override

```
public void moveUp() {  
    center.moveUp();  
}
```

@Override

```
public void moveDown() {  
    center.moveDown();  
}
```

@Override

```
public void moveLeft() {  
    center.moveLeft();  
}
```

@Override

```
public void moveRight() {  
    center.moveRight();  
}
```

@Override

```
public String toString() {  
    return "Circle radius: " + radius + ", center: "  
    + center;  
}
```


Thứ ngày tháng năm

```
public class Main {
```

```
    public static void main (String [] args) {
```

```
        Moveable Point p = new MoveablePoint(0,0,2,3);
```

```
        System.out.println("Trc khi di chuyen: " + p.toString());
```

```
        p.moveUp();
```

```
        p.moveRigth();
```

```
        System.out.println("SauTrc khi di chuyen: " + toString());
```

```
        Moveable c = new MoveableCircle(new  
        MoveablePoint(1, 1, 2, 2), 5);
```

```
        System.out.println("Trc khi di chuyen: "  
        + c.toString());
```

```
Trc
```

```
        c.moveDown();
```

```
        c.moveRight();
```

```
        System.out.println("Sau khi di chuyen: " +  
        c.toString());
```

```
    }
```

```
}
```